

Generation Interconnection System Impact Study Report

For Queue Project Q23-2337

Valley Way 138 kV
September 28, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2337. The Interconnection Customer proposes a 199.9 MW solar photovoltaic facility interconnecting at the Valley Way 138 kV substation in the Pennsylvania area. The request was received on August 04, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 583 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$39,393,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: KNOTT / PRIMM 138 kV line
Contingency: KIRKLAND PARK / ALLEGHENY HYDRO NO 9 138 kV
Loading: 139.2% (Rating: 289 MVA)
DFAX: 5.4%
Pre-project loading: 135.4%
2. Facility: JOHNS CREEK / ELIZABETHTON 138 kV
Contingency: LIONS ROAD / PINEY CRK PROJECT 500 kV
Loading: 135.9% (Rating: 228 MVA)
DFAX: 6.6%
Pre-project loading: 130.1%
3. Facility: TESORO KENAI COGENERATION PLANT / ELIZABETHTON SUBSTATION 138 kV Line
Contingency: LIONS RD SUBSTATION / PINEY CRK PROJECT SUBSTATION 500 kV
Loading: 133.1% (Rating: 236 MVA)
DFAX: 6.6%
Pre-project loading: 127.5%
4. Facility: PRIMM SUBSTATION / JOHNS CRK 138 kV
Contingency: PINEY CREEK PROJECT / LIONS RD 500 kV line
Loading: 131.6% (Rating: 337 MVA)
DFAX: 6.1%
Pre-project loading: 128.0%
5. Facility: TYHEE / STENT TAP 138 kV
Contingency: LIONS RD 345/138 kV TX
Loading: 115.4% (Rating: 207 MVA)
DFAX: 9.7%
Pre-project loading: 106.0%
6. Facility: KNOTT / S SPRINGFIELD 138 kV
Contingency: WARNE TAP / VLY WAY 138 kV
Loading: 113.6% (Rating: 220 MVA)

DFAX: 5.4%

Pre-project loading: 108.7%

7. Facility: ALLEGHENY ENERGY UNITS 3 4 & 5 345-138 kV Transformer

Contingency: ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV XFMR

Loading: 106.3% (Rating: 504 MVA)

DFAX: 6.0%

Pre-project loading: 103.9%

8. Facility: TYHEE / TESORO KENAI COGENERATION PLANT 138 kV

Contingency: LIONS ROAD 345-138 kV TX

Loading: 105.3% (Rating: 221 MVA)

DFAX: 9.7%

Pre-project loading: 96.6%

9. Facility: MERRIAM PARK / VISTA 138 kV L

Contingency: STEWART MOUNTAIN DAM 345/138kV Auto

Loading: 104.4% (Rating: 222 MVA)

DFAX: 7.7%

Pre-project loading: 97.5%

10. Facility: ALLEGHENY HYDRO NO 9 / KIRKLAND PARK 138 kV line

Contingency: VALLEY WAY / WARNE TAP 138 kV

Loading: 104.1% (Rating: 278 MVA)

DFAX: 5.8%

Pre-project loading: 99.9%

11. Facility: YUCCA / FRALEY 138 kV

Contingency: ALLEGHENY ENERGY UNITS 3 4 & 5 / SUGAR HILL 345 kV L

Loading: 103.8% (Rating: 543 MVA)

DFAX: 6.9%

Pre-project loading: 101.2%

12. Facility: FRALEY / HOUSEVILLE 138 kV L

Contingency: SUGAR HILL / ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV

Loading: 103.4% (Rating: 555 MVA)

DFAX: 6.9%

Pre-project loading: 100.9%

13. Facility: VA OSPITAL / HOUSEVILLE 138 kV

Contingency: SUGAR HILL / ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV

Loading: 103.3% (Rating: 560 MVA)

DFAX: 6.9%

Pre-project loading: 100.8%

14. Facility: SUGAR HILL / VA HOSPITAL 138 kV

Contingency: ALLEGHENY ENERGY UNITS 3 4 & 5 / SUGAR HILL 345 kV

Loading: 103.0% (Rating: 570 MVA)

DFAX: 6.9%

Pre-project loading: 100.6%

15. Facility: GUNTERSVILLE DAM / SUGAR HILL 138 kV Line

Contingency: RITTER PARK / SUGAR HILL 345 kV

Loading: 102.6% (Rating: 383 MVA)

DFAX: 14.0%

Pre-project loading: 95.2%

16. Facility: NEW MELONES / HOMER CITY 345 kV L

Contingency: FT GIBSON / ARMSTRONG 500 kV

Loading: 102.0% (Rating: 1,092 MVA)

DFAX: 6.8%

Pre-project loading: 100.7%

17. Facility: BELFIELD / GUNTERSVILLE DAM 138 kV line

Contingency: RITTER PARK / SUGAR HILL 345 kV

Loading: 101.5% (Rating: 415 MVA)

DFAX: 14.0%

Pre-project loading: 94.8%

18. Facility: BELFIELD / MARQUIS 138 kV

Contingency: RITTER PARK / SUGAR HILL 345 kV

Loading: 101.0% (Rating: 433 MVA)

DFAX: 14.0%

Pre-project loading: 94.5%

19. Facility: MARQUIS / FRIENDLY HILLS 138 kV

Contingency: RITTER PARK / SUGAR HILL 345 kV

Loading: 100.5% (Rating: 450 MVA)

DFAX: 14.0%

Pre-project loading: 94.3%

20. Facility: PLACER / KEYSTONE 345 kV

Contingency: SOUTHEAST VICKSBURG / ARMSTRONG 500 kV

Loading: 100.3% (Rating: 1,083 MVA)

DFAX: 7.1%

Pre-project loading: 99.0%

21. Facility: BLOUNTSVILLE / VISTA 138 kV

Contingency: STEWART MOUNTAIN DAM 345/138 kV Autotransformer

Loading: 100.2% (Rating: 245 MVA)

DFAX: 7.7%

Pre-project loading: 93.9%

22. Facility: FIENDLY HILLS / UC DAVIS 138 kV Line

Contingency: SUGAR HILL / RITTER PARK 345 kV line

Loading: 100.1% (Rating: 468 MVA)

DFAX: 14.0%

Pre-project loading: 94.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KNOTT / PRIMM 138 kV	Rebuild line with higher-capacity conductor	\$3,415,000	30% (\$1,033,000)
JOHNS CREEK / ELIZABETHTON 138 kV	Rebuild line with higher-capacity conductor	\$2,628,000	60% (\$1,588,000)
ELIZABETHTON / TESORO KENAI COGENERATION PLANT 138 kV	Rebuild line with higher-capacity conductor	\$7,294,000	60% (\$4,409,000)
PRIMM / JOHNS CRK 138 kV	Rebuild line with higher-capacity conductor	\$5,649,000	35% (\$1,997,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TYHEE / STENT TAP 138 kV	Rebuild line with higher-capacity conductor	\$4,532,000	60% (\$2,743,000)
KNOTT / S SPRINGFIELD 138 kV	Rebuild line with higher-capacity conductor	\$9,570,000	100% (\$9,570,000)
ALLEGHENY ENERGY UNITS 3 4 & 5 345/138kV XFMR	Replace transformer with higher-rated unit	\$12,860,000	43% (\$5,535,000)
TYHEE / TESORO KENAI COGENERATION PLANT 138 kV	Reconductor line	\$5,448,000	60% (\$3,297,000)
MERRIAM PARK / VISTA 138 kV L	Reconductor line	\$322,000	38% (\$121,000)
ALLEGHENY HYDRO NO 9 / KIRKLAND PARK 138 kV line	Reconductor line	\$2,445,000	100% (\$2,445,000)
FRALEY / YUCCA SUBSTATION 138 kV	Reconductor line	\$807,000	28% (\$230,000)
FRALEY SUBSTATION / HOUSEVILLE SUBSTATION 138 kV	Reconductor line	\$1,473,000	28% (\$419,000)
VA HOSPITAL / HOUSEVILLE 138 kV	Reconductor line	\$694,000	28% (\$198,000)
SUGAR HILL / VA HOSPITAL SUBSTATION 138 kV	Reconductor line	\$2,125,000	28% (\$605,000)
GUNTERSVILLE DAM / SUGAR HILL 138 kV	Reconductor line	\$1,093,000	59% (\$648,000)
NEW MELONES / HOMER CITY 345 kV L	Reconductor line	\$8,777,000	5% (\$440,000)
GUNTERSVILLE DAM / BELFIELD 138 kV Line	Reconductor line	\$1,017,000	59% (\$603,000)
BELFIELD / MARQUIS 138 kV	Reconductor line	\$1,116,000	59% (\$661,000)
MARQUIS / FRIENDLY HILLS SUBSTATION 138 kV	Reconductor line	\$1,027,000	59% (\$609,000)
PLACER / KEYSTONE 345 kV	Reconductor line	\$28,853,000	6% (\$1,627,000)
BLOUNTSVILLE / VISTA 138 kV	Reconductor line	\$575,000	38% (\$216,000)
FRIENDLY HILLS / UC DAVIS 138 kV	Reconductor line	\$673,000	59% (\$399,000)
	Total allocated to project		\$39,393,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.